Panskura Banamali College (Autonomous)

B.Sc.(H) 2nd Semester Examination , 2022

Subject : BCA **Paper: GE2P**

Paper Name : Numerical Computing Lab

F.M. : 20 **Time : 2 Hours**

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2020UG009967, paper_code : GE2P , then file name will be :: 2020UG009967- GE2P.pdf for question 'A' and 2020UG009967- GE2P.mp3 for question 'B']

A. Answer any one question: **1 x12 = 12**

1. Write a c program to implement Regula-falsi method.
2. Write a c program to implement Newton- Raphson method.
3. Write a c program to implement Gaussian Elimination
4. Write a c program to implement Gauss-Jordan method.
5. Write a c program to implement Gauss-Jacobi .
6. Write a c program to implement Newtons forward.
7. Write a c program to implement Newtons backward method.
8. Write a c program to implement lagrange's interpolation.

B. Answer any four questions: **4 x2 = 8**

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What is computing ?
2. What is the purpose of having an additional array that carries along the values of an original array?
3. What is the purpose of a Transpose?
4. What form of numerical integration do you think is the most reliable?
5. Why do we bother with Newton Iterations when there are better way to solve things?
6. What is a Taylor Series Expansion?
7. How do you tie the iteration into the DO loop?
8. What is the difference between Newton's method and the Secant method?
9. What does Least Squares do?
10. How exactly do you use the cubic fit and what's its purpose?

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